Example 2.15 A park is circular in shape. The central portion has playthings for kids surrounded by a circular walking pathway. Find the walking area whose outer radius is 10 *m* and inner radius is 3 *m*.

Solution

The radius of the outer circle, $R = 10\ m$

The radius of the inner circle, $r = 3\ m$

The area of the circular path = Area of outer circle – Area of inner circle

$$= \pi R^2 - \pi r^2$$

$$= \pi\left(R^2 - r^2\right)\ sq.units$$

$$= \frac{22}{7} \times (10^2 - 3^2)$$

$$= \frac{22}{7} \times \left((10 \times 10) - (3 \times 3)\right)$$

$$= \frac{22}{7} \times (100 - 9)$$

$$= \frac{22}{7} \times 91$$

$$= 286\ m^2$$

Z5P7S6

Try these

(i) If the outer radius and inner radius of the circles are respectively 9 *cm* and 6 *cm*, find the width of the circular pathway.

(ii) If the area of the circular pathway is 352 *sq.cm* and the outer radius is 16 *cm*, find the inner radius.

(iii) If the area of the inner rectangular region is 15 *sq.cm* and the area covered by the outer rectangular region is 48 *sq.cm*, find the area of the rectangular pathway.

Example 2.16 The radius of a circular flower garden is 21 *m*. A circular path of 14 *m* wide is laid around the garden. Find the area of the circular path.



Fig. 2.24

Solution

The radius of the inner circle $r = 21\ m$

The path is around the inner circle.

Therefore, the radius of the outer circle, $R = 21 + 14 = 35\ m$

The area of the circular path $= \pi\left(R^2 - r^2\right)\ sq.units$

$$= \frac{22}{7} \times (35^2 - 21^2)$$

$$= \frac{22}{7} \times \left((35 \times 35) - (21 \times 21)\right)$$

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